SLIDE POTENTIOMETER EXERCISE

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Map the analogue input reading from a slide potentiometer to a suitable range and use it to control LED brightness via pulse width modulation (PWM).	☐
2	Explain the relationship between the resolution of the analogue input (10-bit ADC) and the resolution of the PWM output (8-bit).	☐

2.0 Hardware Set-Up

This exercise involves connecting an LED to a PWM output pin on the Arduino and controlling its brightness using the 10-bit analogue input from a slide potentiometer, as shown in Figure 1. The potentiometer provides 1024 input levels (0–1023), which are scaled to the 256 PWM levels (0–255) available to drive the LED.

Note: While the ACE-Box can be used for this exercise, it is not required. Only the individual components listed below are needed.

Required hardware

- ATmega328P microcontroller (Arduino Uno)
- USB Type-B cable
- Breadboard
- LED
- 220 Ω resistor
- Slide potentiometer
- 5 × male-to-male breadboard wires

Hardware Assembly Steps

1. Place the LED on the breadboard, connect its cathode to GND, and connect its anode through a 220 Ω resistor to Pin 9.

2. Place the slide potentiometer on the breadboard, connect its outer pins to 5 V and GND, and connect its middle pin (wiper) to A0. Potentiometers vary in type, so check the datasheet for your component to confirm how its three pins map.

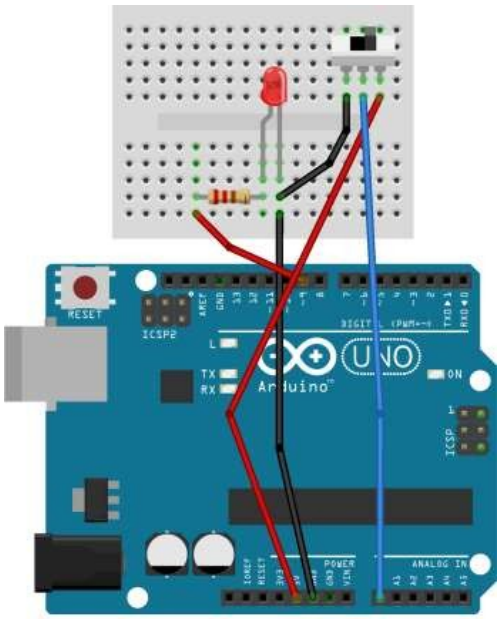


Figure 1: *Hardware Set-Up for the Slide Potentiometer Exercise*

3.0 Simulink Set-Up and Slide Potentiometer Testing

In this part of the exercise, you will develop a Simulink model that controls the brightness of an LED using PWM, based on the analogue reading from the slide potentiometer: as the reading increases, so does the brightness of the LED, as shown in Figure 2. The steps are very similar to those used in the LDR (Light Dependent Resistor) Exercise.

Simulink Model Construction Steps

1. Add an Analog Input block, set the Pin number to A0 and the Sample time to 0.01 s. This reads the sensor voltage and outputs a value between 0 and 1023.
2. Insert a Gain block with a value of 255/1023. This scales the ADC reading into the 0–255 range needed for the PWM duty cycle.
3. Connect the scaled signal to a PWM block and set the Pin number to 9. The PWM block uses the 0–255 value to generate a proportional PWM signal on the Arduino.
4. Connect a Scope block to the signal after the Gain block, so that the scaled PWM value can be viewed in real time as the model runs and the conversion can be verified.
5. Deploy the model to the Arduino. Move the slide potentiometer and observe the PWM value and the LED brightness change.

To recap how to run Simulink code generation and deploy the model to the Arduino Uno, see Section 4.0 of the LED (Digital) Exercise or visit Code Generation.

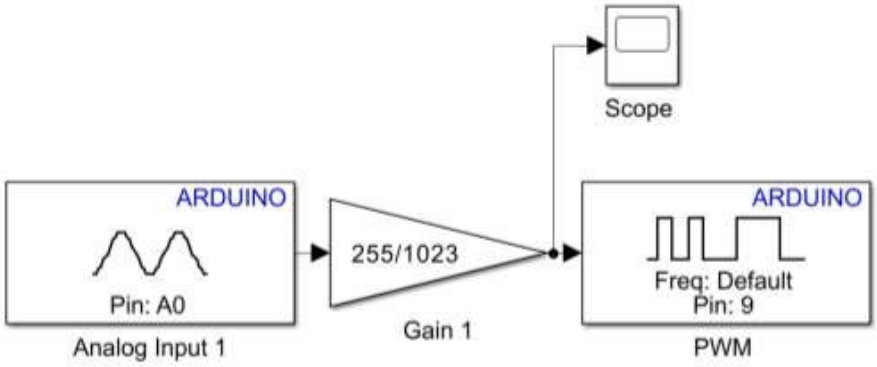


Figure 2: *Simulink Set-Up for the Slide Potentiometer Exercise*

4.0 Exercises

1. Modify the model so that increasing the potentiometer position decreases the LED brightness.
2. Introduce a small dead-band around the mid-position so that small movements do not change the brightness.
3. Add a second LED that switches on when the potentiometer exceeds 75% of its range.

5.0 Concluding Remarks

This exercise has demonstrated analogue signal acquisition, scaling and PWM-based actuator control using Simulink and Arduino hardware. It reinforces the practical relationship between sensor resolution (10-bit ADC) and actuator resolution (8-bit PWM).

You have implemented a complete signal-conditioning chain: acquisition → scaling → actuation → verification. This mapping process is fundamental in embedded control systems, where sensor signals must be transformed appropriately before driving actuators.

Before completing the exercise, return to the Learning Outcomes in Table 1 and confirm that you are confident you can demonstrate each one. If so, tick the corresponding Achieved box.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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